

120 Days of Quantitative Finance

Month 4: Stochastic Calculus & Time Series Modelling

Week 15: Advanced Time Series Modelling

Day 101: AR, MA, and ARMA Models

1 Introduction

Autoregressive (AR) and Moving Average (MA) models are fundamental tools for modeling time series data.

They capture temporal dependencies and are widely used in financial modeling and forecasting.

2 Autoregressive Model AR(p)

An AR(p) model expresses the current value as a linear combination of past values:

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \epsilon_t$$

where:

- ϕ_i are autoregressive coefficients
- $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$ is white noise

2.1 Interpretation

- The process depends on its own history

- Captures persistence or momentum

3 Moving Average Model MA(q)

An MA(q) model expresses the current value as a function of past shocks:

$$X_t = \mu + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t$$

3.1 Interpretation

- Depends on past errors
- Captures short-term shock effects

4 ARMA(p,q) Model

The ARMA model combines both:

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t$$

5 Stationarity Condition

AR models require:

$$|\phi_i| < 1 \quad (\text{for stability})$$

ensuring the process remains stationary.

6 ACF and PACF Behavior

- AR(p): PACF cuts off after lag p , ACF decays
- MA(q): ACF cuts off after lag q , PACF decays
- ARMA: both ACF and PACF decay

7 Financial Interpretation

- AR models capture momentum or mean reversion
- MA models capture reaction to shocks
- ARMA combines both dynamics

8 Conclusion

AR, MA, and ARMA models provide a structured way to model time dependence in financial data and form the basis for more advanced models like ARIMA and GARCH.